

Kalman Filter Procedure

November 28, 2025

Contents

1	Introduction	1
2	Algorithm review	1
3	Test procedure	2
4	Offline processing and analysis	2
5	Performance analysis	2
6	Documentation	2

1 Introduction

A 1D Extended Kalman Filter (EKF) has been developed, with a state vector of altitude, vertical velocity, and a quaternion (4D orientation). It has two phases:

1. **Calibration:** determines initial orientation from gravity to set the quaternion.
2. **Operational:** integrates acceleration to track the system state.

Divergence likely originates in the calibration phase due to an incorrect initial quaternion (then removing gravity in the 'wrong' direction. Resolving this is the first priority.

Once fixed, focus should shift to methodological rigor: structured tests, tuning of Q and R , offline analysis, and robustness measures. This ensures the EKF evolves from a prototype to a reliable engineering solution.

2 Algorithm review

Divergence seems linked to calibration errors. Testing the board with the software is recommended.

3 Test procedure

Define and perform tests to validate the filter and optimize parameters:

1. **Offline (simulated data in MATLAB):**
 - (a) Pure noise
 - (b) Step response
 - (c) Slow ramps
 - (d) Simulated flight (Add sensor noise and physics noise)
 - (e) Real sensor data
2. **Manual motion experiment**
3. **Drone flight test**
4. **Launch test**

During tests, record:

- Sensor outputs
- EKF state vector

4 Offline processing and analysis

Use MATLAB or Python to:

- Visualize filter estimates vs. ground truth
- Compare parameter configurations
- Study gain K , covariance, and residuals

5 Performance analysis

On embedded hardware:

- Measure computation time per prediction-correction cycle

6 Documentation

Clarify:

- Model and noise assumptions
- EKF estimated accuracy